

Architecture Selection & Justification

Multi-Cloud Asset Cost Optimization Portal

1. Architectural Choice

We chose **Layered Architecture** for the Multi-Cloud Asset Cost Optimization Portal. The architecture separates presentation, business processing, and data responsibilities into distinct layers.

2. Reason 1 — Separation of Responsibilities

The portal performs billing-data upload and processing, cost analysis, optimization, dashboard generation, and budget alerts. Layered Architecture keeps these responsibilities separated, making the system easier to understand, maintain, and modify.

3. Reason 2 — Suitable for the Portal Workflow

The portal naturally follows a flow from the web interface to business processing and then to stored billing and cost data. A layered structure matches this workflow and provides clear communication between the major components.

4. Security Advantage

Authentication and authorization can be handled before users access protected cost information and business functions. This supports the requirement that only authorized users can access cloud cost information.

5. Performance Benefit

Separating billing-data processing and database operations from the presentation layer keeps data-management work separate from the user interface and supports efficient processing and dashboard updates for large billing datasets.

Component Summary

Component	Main Responsibility
Web Frontend	User interaction, billing upload, dashboard display
Authentication & Authorization	User authentication and access control
Billing Data Processing	Validate and process cloud billing data
Cost Analysis & Optimization	Categorize costs, identify optimization opportunities and savings
Dashboard & Budget Alerts	Display cost information and generate budget alerts
Database	Store billing, cost, user, and related system data